

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 79 | Date: 13-03-2026

Total Questions: 23 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. EXAM: Mid Term Examination | Set A | Semester 3 --- Section A: Multiple Choice Questions [2 Marks]
2. According to the provided text, what is an architecture fundamentally a result of? a) Only technical decisions made by the development team. b) A set of business and technical decisions. c) The wishes of stakeholders and end-users. d) Primarily the technical environment and available tools. [2 Marks]
3. Which of the following properties is explicitly listed as desirable for having an acceptable system, influencing architects? a) Marketing potential b) Programmer efficiency c) Platform compatibility d) Code complexity [2 Marks]
4. Based on the definition provided in the text, what is an Architectural Pattern? a) An architecture that has been proven to be good. b) A standard decomposition of a known problem into parts. c) A description of element and relation types together with a set of constraints on how they may be used. d) A mapping of functionality onto software elements. [2 Marks]
5. What is a key distinction between an architectural pattern and an architecture, as stated in the text? a) An architectural pattern defines a single, complete system, whereas an architecture is a family of systems. b) An architectural pattern is an architecture, but with additional quality attributes. c) An architectural pattern is not an architecture, but rather defines a set or family of architectures. d) An architectural pattern focuses on runtime elements, while an architecture focuses on static elements. [2 Marks]
6. According to the text, why must architects identify and actively engage stakeholders early in the life cycle? a) To ensure all functional requirements are documented for legal compliance. b) To exclusively focus on managing their technical expectations and negotiating system priorities. c) To understand real system constraints, manage expectations, negotiate priorities, and make tradeoffs. d) To delegate design decisions and avoid personal responsibility for system failures. --- Section B: Short Answer Questions [2 Marks]

7. List and briefly describe two "externally visible" properties of software elements as defined in the text, excluding 'provided services'. [2 Marks]

8. Fault handling: This refers to assumptions other elements can make regarding how a particular element manages or reacts to faults, indicating its ability to correct or mask problems. (Other valid answers from the text include: shared resource usage, and so on.) [5 Marks]
9. Immediate business investment: This includes existing architectural assets, such as an "existing architecture," that the organization has already invested in and may wish to leverage or integrate into the new system. [2 Marks]
10. Describe two advantages of using an Object-Oriented Architectural Style, as presented in the provided text. [2 Marks]
11. Ability to manage errors (Robustness): It provides the capability to effectively manage errors that occur during the execution of the system, enhancing its robustness. (Other valid advantages from the text include: mapping the application to real-world objects for better understanding, providing reusability through polymorphism and abstraction, ability to extend new functionality without affecting the system, improving testability through encapsulation, and reducing development time and cost.) [5 Marks]
12. Explain the six parts of a "Quality attribute scenario" and briefly distinguish between "general" and "concrete" quality attribute scenarios as defined in the text. A "Quality attribute scenario" is a quality-attribute-specific requirement composed of six parts: [2 Marks]
13. Stimulus: A condition that arrives at the system, such as a fault or a stated intention by an actor. [2 Marks]
14. Artifact: The specific part or parts of the system that are stimulated. [2 Marks]
15. Response Measure: How the system's response can be measured and tested. The text distinguishes between: General (quality attribute) scenarios: These are system-independent scenarios where the range of possible values for each of the six parts is typically shown. Concrete (quality attribute) scenarios: These are system-specific scenarios that detail the particular requirements for a given system, making them specific to a particular project or product. [5 Marks]

SECTION B: SHORT ANSWERS

16. Performance characteristics: These are assumptions other elements can make about an element's performance, such as its response time, throughput, or utilization. [5 Marks]
17. Explain how "The Developing Organization" influences an architecture, providing two specific examples mentioned in the text. Two specific examples of this influence are: [5 Marks]
18. Organizational structure: This encompasses factors like whether "subcontracting" is utilized for certain parts of the development, or the "skills of employees" available within the organization, which can dictate the feasibility and design of architectural elements. (Another valid example is "long-term business investment," such as "long term infrastructure.") [5 Marks]

19. Improved maintainability and quality: This style inherently leads to systems that are easier to maintain, and it improves the overall quality of the system primarily due to the promotion of program reuse. [5 Marks]
20. Differentiate between a "Module View" and a "Component-and-Connector (C&C) View" by explaining their primary elements and providing one example property for each, as described in the text. Module Views: Primary Elements: A Module View represents elements that are "implementation units" providing a coherent set of responsibilities. These units might be a class, a collection of classes, a layer, an aspect, or any decomposition of the implementation unit. Example Property: An example property for a module is its Name, which is the primary means to refer to it and often suggests its role in the system. Other properties include responsibilities, visibility of interfaces, and implementation information. Component-and-Connector (C&C) Views: Primary Elements: C&C views show elements that have some "runtime presence." These include "components" (such as processes, objects, clients, servers, and data stores) and "connectors" (which represent pathways of interaction like communication links, protocols, or information flows). Example Property: An example property for an element (component or connector) in a C&C view is Performance, which describes characteristics such as response time under various loads, bandwidth, latency, or throughput, often used for system-wide performance analysis. Other properties include reliability, resource requirements, functionality, security, and concurrency. [5 Marks]
21. Source of Stimulus: The entity that generates the stimulus, which could be an actor, an actuator, or a sensor. [5 Marks]
22. Environment: The conditions surrounding the stimulus, which might be normal operation, degraded operation, or an overload situation. [5 Marks]
23. Response: The action or reaction that the system takes in response to the stimulus. [5 Marks]

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